

INTERNSHIP PROJECT REPORT

Task 4

Binary Classification using Logistic Regression and Decision Tree Classifier

Name: CHANDI PRASAD PADHI

College: ITER

University: Siksha 'O' Anusandhan University

Branch: B.Tech (CSE)

Semester: 2nd Year / 4th Semester

Internship: MainCrafts Technology

Submission Date: 17th Aug 2026

Objective, Dataset and Exploration Objective:

To build and compare classification models using Logistic Regression and Decision Tree Classifier, evaluate performance using classification metrics, analyze the confusion matrix, ROC curve and AUC, and understand class imbalance and class weighting.

Dataset:

The project uses the Breast Cancer dataset from scikit-learn. It contains **569 samples, 30 numerical features** and one binary target variable. The target distribution is 357 samples in class 1 and 212 samples in class 0.

Libraries and Workflow:

Libraries used include pandas, NumPy, Matplotlib, Seaborn and scikit-learn. The workflow includes dataset exploration, an 80:20 stratified train-test split, feature scaling, Logistic Regression, classification metrics, confusion matrix, ROC/AUC analysis, 5-fold cross-validation, class weighting, Decision Tree training and GridSearchCV tuning.

Class Distribution:

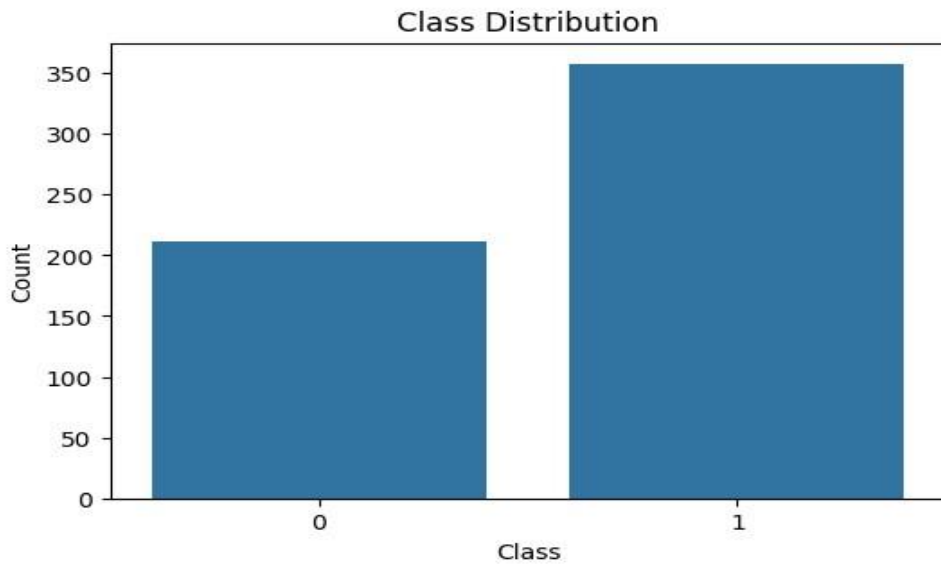


Figure 1. Class distribution of the target variable.

The class distribution shows some imbalance, which motivates the experiment with class-weighted Logistic Regression.

Logistic Regression Evaluation:

The Logistic Regression model was trained after StandardScaler feature scaling. On the test set it achieved **98.25% accuracy**, **98.61% precision**, **98.61% recall** and **98.61% F1-score**.

Confusion Matrix:

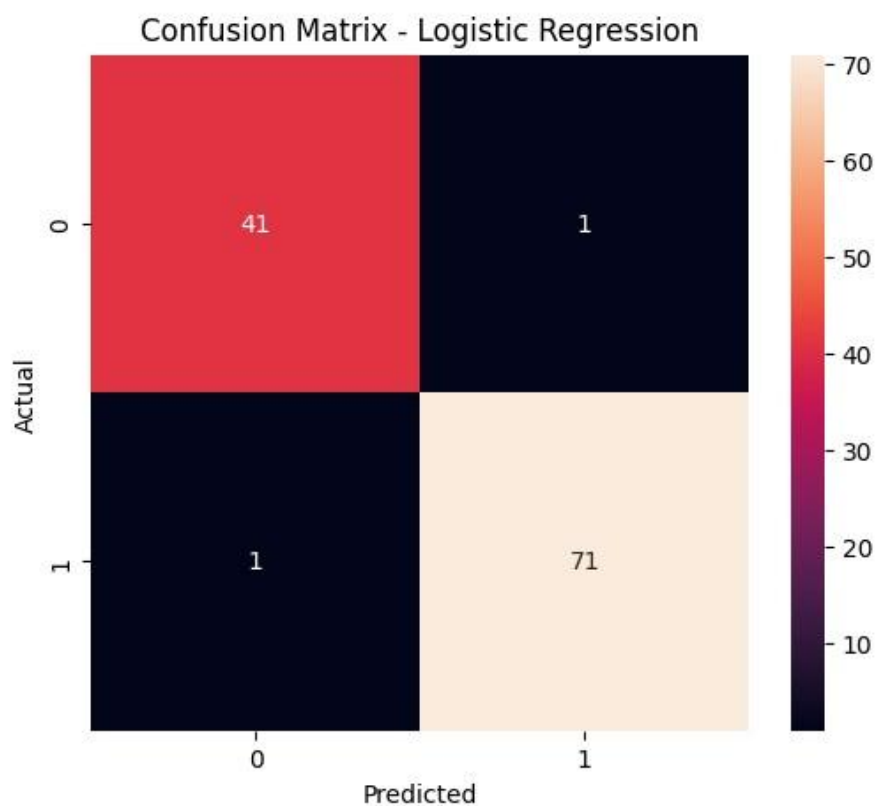


Figure 2. Confusion matrix for Logistic Regression.

The confusion matrix was $[[41, 1], [1, 71]]$, showing only two incorrect predictions out of 114 test samples.

ROC Curve and AUC:

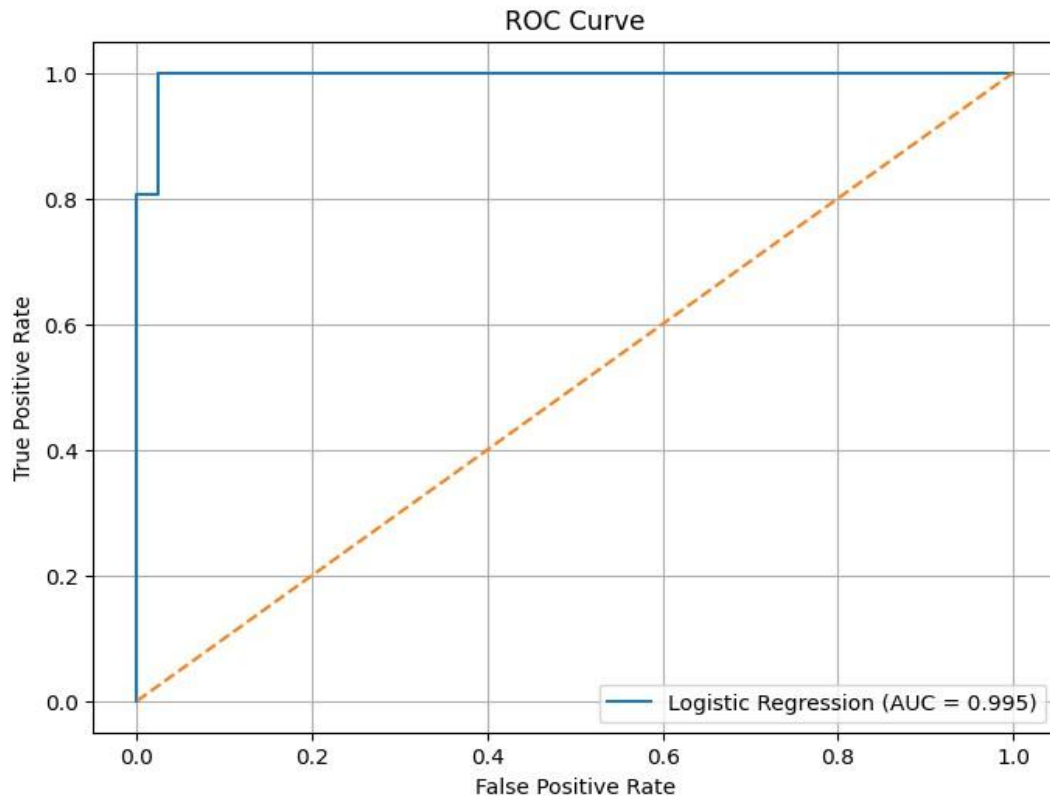


Figure 3. ROC curve for Logistic Regression.

The ROC-AUC score was **0.99537**, indicating excellent discrimination between the two classes. The 5-fold cross-validation average F1-score was **0.98434**.

Model Comparison, Tuning and Final Result Handling Class Imbalance:

A class-weighted Logistic Regression model was tested using `class_weight='balanced'`. It achieved Accuracy 0.95614, Precision 0.98551, Recall 0.94444 and F1-score 0.96454.

Decision Tree and Hyperparameter Tuning:

The initial Decision Tree achieved Accuracy 0.91228, Precision 0.95588, Recall 0.90278 and F1-score 0.92857. GridSearchCV with

5-fold cross-validation searched over `max_depth` and `min_samples_split`. The best parameters were **`max_depth = 5`** and **`min_samples_split = 5`**, with best CV F1-score **0.95177**. The tuned Decision Tree achieved Accuracy 0.92105, Precision 0.95652, Recall 0.91667 and F1-score 0.93617.

Final Model Comparison:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.9825	0.9861	0.9861	0.9861
Balanced Logistic Regression	0.9561	0.9855	0.9444	0.9645
Decision Tree	0.9123	0.9559	0.9028	0.9286
Tuned Decision Tree	0.9211	0.9565	0.9167	0.9362

Best Model:

Based on the highest F1-score, the notebook selects **Logistic Regression** as the best model. It achieved **98.25% accuracy** and **98.61% F1-score** on the test set.

Conclusion:

This task demonstrates a complete binary classification workflow: data exploration, preprocessing, model training, evaluation with accuracy/precision/recall/F1, confusion-matrix analysis, ROC-AUC evaluation, cross-validation, class weighting, Decision Tree comparison and GridSearchCV hyperparameter tuning. The final model selection is based particularly on F1-score.